

📘 AI/ML Internship – Day 25 Notes & Tasks

🟡 Module 4: NumPy (Real-World Problems & Practice)

📌 Part 1: Why Practice Real Problems?

👉 In AI/ML:

- Real datasets are messy
- Problems require logic + NumPy
- Multiple concepts work together

👉 Today combines:

- Arrays
 - Slicing
 - Broadcasting
 - Aggregation
 - Reshaping
-

📌 Part 2: Theory Answers (IMPORTANT)

◆ 1. Why NumPy is Important in AI/ML?

👉 NumPy helps:

- Fast computations
 - Data handling
 - Matrix operations
 - Model calculations
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◆ 2. What is Vectorized Computation?

👉 Performing operations on complete arrays without loops.

👉 Faster and efficient.

♦ 3. Why Shape is Important?

👉 ML models expect specific input shapes.

Incorrect shape → errors.

♦ 4. What is Feature Extraction?

👉 Selecting useful information from data.

Example:

- Average marks
 - Pixel intensity
-

♦ 5. Why NumPy is Faster than Python Lists?

👉 NumPy:

- Uses optimized C implementation
 - Uses contiguous memory
 - Supports vectorized operations
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🔴 Part 3: Problem 1 – Student Marks Analysis

```
import numpy as np
```

```
marks = np.array([
    [85, 78, 90],
    [70, 88, 60],
    [95, 92, 89]
])
```

```
print("Average:", np.mean(marks))
```

```
print("Highest:", np.max(marks))  
print("Lowest:", np.min(marks))
```

🔴 Part 4: Problem 2 – Find Top Student

```
total = np.sum(marks, axis=1)  
  
print(total)  
  
top_student = np.argmax(total)  
  
print("Top Student Index:", top_student)
```

🔴 Part 5: Problem 3 – Image Brightness Increase

```
image = np.array([  
    [50, 100],  
    [150, 200]  
])
```

```
bright = image + 50
```

```
print(bright)
```

👉 Broadcasting used automatically.

🔴 Part 6: Problem 4 – Normalize Dataset

- ◆ Why Normalize?

👉 Bring values into smaller range.

```
data = np.array([10, 20, 30, 40])
```

```
normalized = data / np.max(data)
```

```
print(normalized)
```

📌 Part 7: Problem 5 – Filter Data

```
arr = np.array([10, 25, 30, 5, 60])
```

```
filtered = arr[arr > 20]
```

```
print(filtered)
```

📌 Part 8: Problem 6 – Reshape ML Input

```
arr = np.arange(12)
```

```
reshaped = arr.reshape(3, 4)
```

```
print(reshaped)
```

📌 Part 9: Problem 7 – Simulate Attendance Dataset

```
attendance = np.random.randint(0, 2, size=(5, 7))
```

```
print(attendance)
```

👉 Meaning:

- 5 students
 - 7 days
 - 0 = absent
 - 1 = present
-

📌 Part 10: Real-World Thinking

👉 In real AI:

- Images = arrays
- Datasets = matrices
- Features = numerical values

👉 NumPy is base for:

- Pandas
- TensorFlow
- PyTorch
- Scikit-learn

📌 Part 11: Visualization of Dataset

Student Marks Matrix

```
[[85 78 90]  
 [70 88 60]  
 [95 92 89]]
```

Rows → Students

Columns → Subjects

🎯 Tasks for Day 25

🧠 Task 1: Theory (Write Answers)

1. Why NumPy is important?
 2. What is vectorization?
 3. Why shape matters in ML?
 4. What is normalization?
 5. What is feature extraction?
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Task 2: Marks Analysis

1. Find average marks
 2. Find highest marks
 3. Find lowest marks
 4. Find topper
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Task 3: Image Operations

1. Increase brightness
 2. Decrease brightness
 3. Normalize image pixels
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Task 4: Dataset Filtering

1. Filter values > 50
 2. Replace low values
 3. Count filtered values
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Task 5: Reshaping

1. Convert 1D → 2D
 2. Convert 2D → 1D
 3. Create 3D dataset
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Task 6: Dataset Simulation

1. Attendance matrix
2. Marks dataset
3. Weather dataset